

OPERATION BLACK VAULT

The Vanishing Profit

End-to-End Business Intelligence Case Study

PROJECT STACK

Excel → MySQL → Power Query → Power BI → DAX → Business Insights

Global Superstore Retail Analytics

[INSERT 3-DASHBOARD COLLAGE SCREENSHOT HERE]

Prepared as a portfolio-ready Data Analytics project • 2026

2. Executive Summary

From raw transactions to decision-ready business intelligence

THE CASE IN ONE MINUTE

Operation Black Vault transforms retail transaction data into an interactive analytical solution. The project validates data with SQL, builds a structured Power BI model, creates reusable DAX measures, and converts product, market, customer, and profitability patterns into management-focused insights.

What We Built

A three-page Power BI reporting solution covering Executive Overview, Product & Market Analysis, and Customer & Business Insights.

What the Analysis Answers

- How are sales and profit performing over time?
- Which categories, products, markets, and segments drive business performance?
- Where are low-profit or underperforming areas concentrated?
- Which customer and business factors are associated with profitability?
- What practical actions can management take?

End-to-End Outcome

The final solution combines data preparation, SQL validation, Power Query transformation, star-schema modeling, DAX calculations, interactive visuals, AI-assisted analysis, and a business report.

PORTFOLIO VALUE

This is not only a dashboard project. It demonstrates the complete analyst workflow: understand → validate → transform → model → analyze → communicate → recommend.

3. Business Problem & Analytical Objectives

Business Problem

A retail organization can generate substantial transaction data while still lacking a clear view of where revenue is created, where profit is lost, and which business levers deserve management attention. The challenge is to move beyond descriptive totals and identify the patterns behind performance.

Primary Objective

Build a reliable, interactive Business Intelligence solution that converts transactional data into actionable insight on sales, profitability, products, markets, customers, and business drivers.

Key Business Questions

Area	Question
Performance	How do sales and profit change over time?
Product	Which categories, sub-categories, and products create the most value?
Market	Which markets perform strongly and which require attention?
Customer	Does high sales value always translate into high profitability?
Profitability	Which factors appear to influence profit?
Action	What decisions could improve profitability and resource allocation?

Success Criteria

- Accurate and validated data
- A maintainable analytical model
- Reusable KPI measures
- Clear executive dashboards
- Evidence-based recommendations

4. Dataset Overview & Business Context

The project uses the Global Superstore retail dataset, a transactional business dataset commonly used for sales, profitability, product, customer, market, and shipping analysis.

Core Tables

Table	Role	Business Use
Orders	Primary transaction table	Sales, profit, customers, products, dates, geography, shipping
Returns	Return reference	Identify returned orders and support return-related analysis
People	Manager reference	Connect managers to regional/business areas

Key Business Dimensions

Customer • Product • Category • Sub-Category • Market • Region • Country • Segment • Ship Mode • Order Date • Ship Date

Key Measures

Metric	Purpose
Total Sales	Overall revenue generated
Total Profit	Overall profit contribution
Profit Margin	Profitability relative to sales
Total Orders	Transaction/order volume
Total Quantity	Units sold
Average Shipping Days	Operational delivery measure

DATA STORY

The dataset connects financial performance with products, customers, geography, and time—allowing the report to investigate both 'what happened' and 'where to look next'.

5. Project Methodology & Technology Stack

End-to-End Methodology

#	Stage	Outcome
1	Data Understanding	Review structure, fields, business entities, and requirements.
2	Data Preparation	Clean, validate, and transform source data.
3	SQL Validation	Check completeness, duplicates, validity, relationships, and business metrics.
4	Power BI Modeling	Build relationships and reporting-ready model.
5	DAX & Visualization	Create reusable measures, KPIs, trends, rankings, and advanced visuals.
6	Business Storytelling	Interpret findings and translate them into recommendations.

Technology Stack

Tool	Role
Microsoft Excel	Initial exploration and preparation
MySQL	Data import, validation, SQL business analysis
Power Query	Transformation and column/data-type management
Power BI Desktop	Data modeling and interactive reporting
DAX	Business measures and KPI logic
Microsoft Word	Final business report

DESIGN PRINCIPLE

Use the simplest visual that answers the business question. The final dashboards prioritize readability and interview explainability over visual complexity.

6. Data Cleaning, Validation & Data Modeling

Data Quality Controls

Check	Approach	Outcome
Completeness	Missing-value checks	Validated before analysis
Duplicates	Duplicate transaction/order checks	Investigated and validated
Validity	Date, numeric, business-rule checks	Validated
Consistency	Cross-check sales/profit and related fields	Validated
Relationships	Key and relationship checks	Validated
Business validation	SQL totals compared with reporting logic	Control step

Power Query Transformations

- Removed unnecessary columns to reduce model clutter.
- Standardized and verified data types.
- Prepared data for relationship creation.
- Created/used a dedicated date structure for monthly analysis.

Data Model

A structured reporting model was created in Power BI with the transaction table at the center and supporting dimensions/reference tables around it.

[INSERT POWER BI MODEL VIEW SCREENSHOT HERE]

WHY THIS MATTERS

Good visuals cannot compensate for unreliable data or ambiguous relationships. Validation and modeling were prerequisites to dashboard development.

7. SQL Analysis & Business Validation

Role of SQL

MySQL was used as an independent validation and analysis layer before final dashboard storytelling. This reduced the risk of building visual conclusions on incorrect imports, missing values, duplicates, or inconsistent relationships.

Analysis Areas

Analysis	Business Question
Data quality	Is the imported dataset complete and valid?
Product analysis	Which products/categories contribute most to sales and profit?
Market analysis	Which markets/regions perform strongly or weakly?
Time analysis	How does performance change by month/year?
Shipping analysis	What patterns exist in delivery performance?
Customer analysis	Which customers create high value or low profitability?

SQL Techniques Applied

SELECT • WHERE • GROUP BY • HAVING • ORDER BY • Aggregate Functions • JOINS • CASE WHEN • CTEs • Window Functions • Date Functions

[INSERT 1–2 SQL SCREENSHOTS HERE]

ANALYST CONTROL

SQL served as a control layer to verify the numbers later presented in Power BI.

8. Power BI Data Model, DAX & KPI Development

Model-to-Insight Pipeline

Validated data was transformed in Power Query, structured into the reporting model, and then exposed through reusable DAX measures. This separation keeps calculations consistent across visuals and pages.

Core KPI Layer

Measure	Business Meaning
Total Sales	Total revenue in selected context
Total Profit	Total profit in selected context
Profit Margin	Profit divided by sales
Total Orders	Order volume
Total Quantity	Quantity sold
Average Shipping Days	Average delivery duration

Dashboard Design Decisions

- Executive KPIs were kept prominent and easy to scan.
- Monthly sales and profit trends were separated for clarity.
- Slicers were kept simple to preserve dashboard space.
- Product and market pages use ranking and decomposition for deeper analysis.
- Key Influencers and Smart Narrative were used where they added explanatory value.

IMPORTANT CORRECTION

Profit Margin represents profitability, not revenue divided by profit. The business logic is Profit ÷ Sales, formatted as a percentage.

9. Dashboard 1 — Executive Overview

Purpose: provide management with a fast, high-level view of business health and interactive filtering.

[INSERT FINAL EXECUTIVE OVERVIEW SCREENSHOT — LARGE]

Visual Story

Component	Decision Supported
KPI Cards	Immediate view of sales, profit, margin, orders, and volume
Monthly Sales Trend	Identify revenue movement over time
Monthly Profit Trend	Monitor profitability independently from revenue
Category Analysis	Compare major product categories
Segment Analysis	Understand customer-segment contribution
Slicers	Explore KPIs under different filters

Interpretation

The executive page is intentionally concise. It answers: 'How are we doing?', 'Is profit keeping pace?', and 'Where should I investigate next?'

EXECUTIVE USE

Use this page as the starting point for a management review. The deeper pages explain product, market, customer, and profit-driver patterns.

10. Dashboard 2 — Product & Market Analysis

Purpose: move from headline performance into product and market dimensions that explain where value is being created.

[INSERT FINAL PRODUCT & MARKET DASHBOARD SCREENSHOT — LARGE]

Visual Story

Visual	Question It Answers
Treemap	Which categories/sub-categories dominate contribution?
Matrix	How do category and sub-category sales/profit compare?
Top N	Which products deserve attention or replication?
Bottom N	Which products need review?
Market & Profit Bar	Which markets contribute most/least profit?
Decomposition Tree	What dimensions explain a selected profit result?

Business Interpretation

This page turns a large product catalog into a manageable decision set. Top/Bottom analysis supports prioritization, while decomposition helps move from a total profit result toward the dimensions that explain it.

MANAGEMENT QUESTION

Which products and markets deserve more investment—and which should be reviewed for pricing, cost, assortment, or strategy?

11. Dashboard 3 — Customer & Business Insights

Purpose: investigate profitability at a deeper level and explore relationships and potential drivers.

[INSERT FINAL CUSTOMER & BUSINESS INSIGHTS SCREENSHOT — LARGE]

Advanced Analytics

Visual	Analytical Role
Scatter Plot	Explore relationship between customer sales and profit
Key Influencers	Identify factors associated with higher/lower profit
Smart Narrative	Generate an automated summary of selected findings
Ribbon/Ranking Analysis	Observe changes in performance ordering over time
Supporting visuals	Provide context around customer/category performance

Why This Page Matters

High sales does not automatically mean high profitability. This page asks which customers, products, discounts, and business conditions are associated with profitable outcomes.

ANALYTICAL MINDSET

The strongest question is not only "What happened?" but also "Why might it be happening, and what should the business investigate next?"

12. Executive Business Insights

#	Insight	Business Meaning
1	Profitability should be monitored alongside sales.	Revenue growth alone does not establish business health; profit and margin provide the critical second lens.
2	Technology is a major performance driver.	Category analysis identifies Technology as a key area of business contribution.
3	Product performance is uneven.	A smaller group of products can contribute disproportionately to outcomes, making ranking analysis valuable.
4	Markets do not perform uniformly.	Market comparison highlights where performance is strong and where attention may be required.
5	High-sales customers are not automatically the most profitable.	Customer analysis supports a value-based rather than revenue-only view.
6	Discount strategy deserves profitability monitoring.	Discount-related patterns can materially affect profit generated by orders/products.
7	Time trends reveal direction.	Monthly sales and profit trends provide context that a single total cannot.
8	Interactive reporting improves investigation.	Slicers and drillable visuals allow stakeholders to move from summary to detail.

CORE TAKEAWAY

The project shifts the conversation from 'How much did we sell?' to 'Where did value come from, what is reducing profitability, and where should management focus next?'

13. Business Recommendations

Priority	Recommendation	Action	Expected Impact
1	Protect profitable revenue	Review discounts and promotions using profit impact, not sales uplift alone.	Higher-quality revenue and stronger margins.
2	Scale proven product strengths	Prioritize high-performing categories/products for inventory, marketing, and assortment decisions.	Better resource allocation.
3	Review persistent low-profit products	Investigate pricing, cost, discount, and assortment factors.	Reduced profit leakage.
4	Prioritize attractive markets	Use market profitability alongside sales volume when deciding where to invest.	More targeted expansion.
5	Manage customers by value	Combine sales and profit when identifying strategic customers and retention priorities.	Higher customer value.
6	Institutionalize dashboard reviews	Use the report as a recurring management review tool.	Faster, consistent decisions.

Recommendation Framework

Finding → Diagnose → Decide → Monitor. The dashboard should trigger a business question; the recommendation should be validated with operational context before implementation.

IMPORTANT

These are analytical decision directions, not claims that operational changes were implemented. A real business would validate cost, pricing, inventory, and customer context before execution.

14. Challenges, Technical Issues & Solutions

Challenge	Diagnosis	Resolution
CSV/data import issues	Source/import configuration problems	Verified source data, corrected import approach, and revalidated row counts.
MySQL configuration/import errors	LOCAL INFILE / file constraints	Adjusted the import configuration/process and validated resulting tables.
Data quality uncertainty	Need to prove completeness and consistency	Ran targeted SQL checks for missing values, duplicates, validity, consistency, and relationships.
Power BI modeling decisions	Relationship/model ambiguity	Reduced model clutter, created relationships carefully, and structured the model for reporting.
Date/month analysis	Monthly reporting not behaving as expected	Used a proper date structure and configured date-based analysis.
Profit Margin logic	Implausible percentage result	Corrected measure to Profit ÷ Sales and formatted as percentage.
Filled Map/custom visual limitations	Visual dependency was unreliable	Used a more dependable visual approach rather than blocking the report.
KPI/icon layout	Decorative elements moved relative to cards	Reworked positioning/layering and visual layout.
Dashboard density	Advanced visuals risked complexity	Simplified arrangement, spacing, titles, and visual roles.

What We Learned

The strongest learning was the habit of diagnosing the data/model first, validating calculations independently, and simplifying the final presentation around business questions.

15. Conclusion, Future Scope & Portfolio Readiness

Project Conclusion

Operation Black Vault demonstrates an end-to-end analytics workflow: raw retail data was prepared and validated, analyzed with SQL, transformed and modeled in Power BI, quantified through DAX, visualized through three dashboard pages, and translated into business insights and recommendations.

What the Project Demonstrates

- Data preparation and quality validation
- SQL-based analytical thinking
- Power Query transformation
- Power BI data modeling
- DAX KPI development
- Executive dashboard design
- Advanced analytical visuals
- Business storytelling and recommendations
- Practical debugging and problem solving

Future Scope

- Add growth indicators where decision context requires them.
- Introduce forecasting for sales and profit trends.
- Deploy through Power BI Service with governed refresh.
- Explore Row-Level Security for role-based access.
- Add returns/shipping KPIs where validated data supports them.
- Create a documented semantic model and reusable KPI library.

FINAL PORTFOLIO STATEMENT

This project demonstrates analyst capability beyond dashboard construction: reliable data, structured modeling, analytical reasoning, clear communication, and business-focused decision support.

Portfolio Publishing

Publish the project to GitHub with the PBIX/report, SQL scripts, screenshots, business report, and a concise README; then present it on LinkedIn using a short problem → approach → insight → recommendation story.